nTop

NOTEBOOK API, ADDITIVE MANUFACTURING

2D Nesting of Parts

A custom block that packs implicit bodies on the build plate, from inside nTop, November 2026

One block that packs the bodies you give it

Give it implicit bodies, a plate and a folder. It returns the same bodies rotated and moved into a tight nest on the plate.

The packing runs outside nTop, but the block drives it

Each body casts a vertical shadow. The block writes those shadows to a folder, launches the packer, waits for it, reads the placements back and applies them with Transform Object. Nothing is copied by hand and nothing is typed in.

- Packs on the true footprint of each body, holes and concavities included, not on bounding boxes
- Part Spacing of 0 lets shadows touch; any other value is the minimum gap between neighbours
- Rotates about Z only, so every body keeps the build orientation you gave it
- Works for any number of bodies; the chain is list-processed, not copied per part

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Five inputs, one output

INPUTS

What you wire in

- **Bodies**: the implicit bodies to nest, any count
- **Part Spacing**: minimum gap between shadows, 0 = touching
- **Build Plane**: the plate plane, normal up, parallel to world XY
- **Build Plate**: the printer's build volume; the nest starts at its min corner
- **Work Folder**: a writable folder that holds nest_pack.exe; the block writes its files and runs the packer there

OUTPUT AND SETTINGS

What comes out

A list of the same bodies, each rotated about the Z axis through the origin and moved onto the plate. Wire it wherever the original list went.

Settings inside the block

- Part Lift 5 mm, Plate Margin 0 mm, Rotation Step 15°
- Raster Pixels 400, Packer Timeout 900 s, Shadow Clearance 10 mm, Shadow Feature Size 0.5 mm

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What you need on your machine

One install, one folder, one permission. Everything else the block fetches or carries itself.

- **nTop 6.3.3 or newer**: the block travels inside the .ntop file, nothing to register
- **nest_pack.exe in the Work Folder**: self-contained, nothing to download or install, Windows only
- **A writable Work Folder**: the block writes its images and tables there and runs the packer there
- **One permission per Work Folder**: nTop asks whether the notebook may run an external command the first time it sees that command text; allow it
- **No Python of your own and no network**: nothing else is installed or changed on the machine

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Shadow, pack, apply

Seven sections inside the block, each opening with a comment on what happens there and why. Three do the work; the rest carry numbers between nTop and the packer.

1: SHADOWS AND BOUNDS

The footprint of each body

A Manufacturing Support Volume at 90° overhang runs from each body down to a plane 10 mm below the lowest body. Converted back to an implicit, that prism is the body's exact shadow.

- Bounds come from bounding boxes read through three plane fields
- Each body's lowest point sets its lift onto the plate

2: EXPORT AND PACK

Files out, packer in

Every shadow is sliced to an image over its own bounds, and a table lists the bounds and heights. nTop's Run Command block then runs nest_pack.py in the Work Folder.

- The command's timeout is derived from the exports, so it cannot start early
- Bottom-left fill on a 0.5 mm grid, rotation sweep, smallest nest wins

3: IMPORT AND APPLY

Placements become transforms

The packer writes one row per body: angle, dx, dy, dz. The block imports that table, turns the columns into Angles and Moves, and applies them with two Transform Object blocks.

- The result file is named by the exit code, so a failed run never feeds a stale layout
- Rotation about the origin first, then the move

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Why the packer ships as nest_pack.exe

One 33 MB file next to the block. It carries its own Python, numpy and Pillow, so nothing is downloaded or installed on the machine that runs it.

HOW THE BLOCK USES IT

One command, no runtime

nTop's Run Command launches nest_pack.exe from the Work Folder with the folder as its only argument. Everything else it needs is already in that folder.

- Parameters travel in params.csv, so the command text never changes and nTop asks permission once per folder.
- No Python, no package index, no network, no admin rights. Nothing on the machine is changed.
- Built from nest_pack.py with PyInstaller; scripts/build_exe.ps1 rebuilds it after an edit, one build per operating system.

WHY NOT SOMETHING ELSE

The alternatives cost more

- A system Python needs the right version and two packages installed by hand on every machine, and breaks when either changes.
- A package manager such as uv fetches Python and the packages itself, but that is a download from a public index, which some customers cannot allow.
- nTop's own console Python has no numpy or Pillow and cannot be launched by a block.
- The exe trades 33 MB and a rebuild per edit for zero setup. Code-sign it before release so corporate antivirus accepts it.

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What nest_pack.py does

About 280 lines of numpy and Pillow, shipped as nest_pack.exe beside the block; the source stays readable in nest_pack.py.

■ 1. START AND READ

Start and read

nest_pack.exe gets the folder as its only argument and reads spacing, lift, rotation step, plate bounds and margin from params.csv, so the command nTop runs is the same every time.

- parts.csv gives each shadow's bounds and each body's height
- Each shadow image is resampled onto a 0.5 mm grid, coarser when the largest part passes about 110 mm

■ 2. PACK

Bottom-left fill on a grid

Each shadow becomes a boolean mask, rotated in Rotation Step increments. A FFT correlation of the plate occupancy with the mask finds every position where the part fits in one pass.

- Placed parts are dilated by the spacing; the position that keeps the nest's bounding box smallest wins
- Every part order is tried up to four parts; beyond that a few orders under a four-minute budget

■ 3. WRITE BACK

One row per body

placements0.csv holds angle, dx, dy and dz for each body in input order. The rotation found about the shadow's centre is re-expressed about the world origin so nTop needs only one rotation and one move.

- nest_pack.log records what was read, the nest size and every placement
- Parts that fit nowhere are parked in a row beside the plate, flagged placed = 0 and listed in unplaced.csv for a second plate

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What the block leaves on disk

Everything the block and the packer exchange is a plain file in the Work Folder, so a run can be inspected after the fact.

WRITTEN BY THE BLOCK

Inputs to the packer

- **shadow_0, shadow_1, ...**: one image archive per body, white inside, over that body's own bounds
- **parts.csv**: per body: minx, miny, maxx, maxy of the shadow and minz, maxz of the body
- **params.csv**: spacing, lift, rotation step, plate bounds and margin, so the command line itself never changes

WRITTEN BY THE PACKER

Results back to nTop

- **placements0.csv and unplaced.csv**: angle_deg, dx, dy, dz and a placed flag per body, in input order, plus the indices of parts parked beside the plate
- **nest_pack.log**: what the packer read, the nest size and every placement
- **nest_pack.exe**: the shipped executable you copy into the folder once; nest_pack.py beside the block is its readable source

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Six steps to a nest

The first run takes about a minute for a handful of bodies; later runs only repeat when an input changes.

- **1. Pick a writable Work Folder** and copy nest_pack.exe into it; the block fills in the rest
- **2. Place 2D Nesting of Parts** from the custom blocks panel, or embed 2D Nesting of Parts.ntop into a new notebook
- **3. Wire the bodies** as one implicit list, in the order you want to read the results back
- **4. Wire the printer**: the build volume box and its floor plane, for example from the EOS M 280 toolkit block
- **5. Set Part Spacing**: 0 for touching shadows, or the gap your process needs
- **6. Build**: the first run may ask permission to run an external command; allow it once and the block does the rest

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Limits and troubleshooting

WHAT IT ASSUMES

Scope of the first version

- The Build Plane is parallel to world XY; bounds are read along the world axes
- Bodies may sit anywhere in Z, even below the plate; the shadow plane follows the lowest one
- Packing is on a 0.5 mm raster (coarser when the largest part exceeds 110 mm), so touching parts keep a small gap
- Up to four bodies every order is tried; more bodies get a few orders and a 4-minute budget, and the raster coarsens for large parts

WHEN SOMETHING IS OFF

Where to look first

- Parts that do not fit are parked in a row beside the plate's +X edge and listed in unplaced.csv; the rest still nest
- Packer Run stays unbuilt: toggle one of its inputs; a command block does not re-run on its own after a recipe import
- Parts off the plate: check Build Plate and Build Plane come from the same printer block
- Odd rotations: angles must carry degree units; a unitless value is read as radians

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Thank you

2D Nesting of Parts.ntop ships with nest_pack.exe

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